Does Knowing Your Opponent Help? Planning and Plan Recognition in Four-Player Backgammon

Elena Marwick

Department of Computer Science, Northvale Institute of Technology
1200 Harbor Road, Northvale, CA 95400, USA
emarwick@cs.northvale-tech.edu

Abstract

Backgammon has served as an AI benchmark for decades, but almost exclusively in its two-player form. With more than two players the game stops being a zero-sum contest between two sides, and modelling *individual* opponents becomes potentially more valuable. We introduce a free-for-all four-player backgammon variant that stays as close as possible to the standard rules, and release it as an open-source testbed. On top of it we build an agent that combines selective expectimax lookahead with Bayesian plan recognition: each opponent is modelled as a Boltzmann-rational follower of one hidden plan from a library of five playing styles plus a random plan, and the posterior over plans is used to predict that opponent's replies inside the search. Lookahead significantly improves on the greedy evaluation it is built on. Plan recognition is accurate (81.9% top-1 accuracy against 16.7% chance, 69% after only ten observed decisions), yet across four experiments it produces no statistically significant change in playing strength. Decision logging shows why: recognition changes only 2.8–5.6% of the planner's moves, and a planner that is *told* each opponent's true plan does no better. In our setting the bottleneck is how much the planner can exploit an opponent model, not how accurate that model is. We argue that this is a general lesson for opponent modelling in multi-player stochastic games.

1 Introduction

Backgammon occupies a special place in the history of game-playing AI. BKG 9.8 was the first program to defeat a reigning world champion in any board game (Berliner 1980), TD-Gammon showed that self-play temporal difference learning could reach master level (Tesauro 1994, 1995), and the game is still used as a benchmark for planning under chance, for example by Stochastic MuZero (Antonoglou et al. 2022). Almost all of this work, however, concerns the two-player game. Two-player backgammon is zero-sum: what is good for one side is bad for the other, and there is no need to distinguish one opponent from another.

With more players this changes. A player at a four-player table faces several opponents with their own goals, it interacts with each of them differently, and its final result depends on all of these interactions. Whether a neighbour races, attacks, or builds a blockade determines which of one's own checkers are in danger and which positions are safe. Modelling individual opponents therefore seems more valuable than in the two-player game. At the same time, the classical multi-player search algorithms ($\max^n$, paranoid, best-reply search) must make an assumption about how opponents behave, and the right assumption is rarely obvious (Sturtevant 2002; Nijssen and Winands 2012).

In this paper we ask a simple question: *if an agent can work out how each of its opponents plays, does it play better?* To study this question we make three contributions.

1. **A four-player backgammon variant and open-source testbed.** There is no standard free-for-all four-player backgammon, and we found no academic AI work on one. We define a variant on a circular 48-point board that keeps the standard setup, movement, hitting and bearing-off rules, and in which each player interacts with both of its neighbours. We release a dependency-free engine with reproducible, seat-rotated experiment infrastructure.
2. **A planning agent with plan recognition as its opponent model.** The agent keeps a Bayesian posterior over which plan from a library of playing styles each opponent follows. The observation model is the Boltzmann-rational likelihood used in goal recognition as planning and Bayesian inverse planning (Ramírez and Geffner 2010; Baker, Saxe, and Tenenbaum 2009), applied to evaluation-function values instead of plan costs. The posterior-weighted mixture of plan policies predicts opponent replies inside a selective expectimax lookahead over the dice.
3. **An empirical study that separates recognition from exploitation.** We measure recognition accuracy on its own, playing strength with and without recognition (ablation), and, in an analysis added for this paper, how many decisions recognition changes and how a planner given the *true* opponent plans (oracle) performs. Recognition is fast and accurate, but it does not lead to a significant gain in playing strength, and neither does perfect knowledge of opponent plans.

The negative result is a central message of this paper. Accurate opponent models are often treated as useful in themselves, and plan recognition is rarely evaluated *inside* a planner by its effect on results. Our experiments show a case where a well-founded and accurate recognizer adds nothing measurable, and they locate the reason: the planner only acts on the model through a narrow channel, the risk of being hit,

and within that channel the plans that are hard to tell apart also lead to nearly the same predictions. This agrees with recent observations that in multi-player games an agent's own moves often matter more than accurate predictions of its opponents (Baier and Kaisers 2020), and it gives concrete guidance on where opponent models need to enter a planner in order to pay off.

2 Related Work

Backgammon AI. After the hand-crafted evaluation of BKG (Berliner 1980), neural evaluations trained by self-play came to dominate (Tesauro 1995, 2002; Pollack and Blair 1998). Search in backgammon must handle chance nodes; expectimax (Michie 1966) and *-minimax (Ballard 1983) are the standard tools, and Hauk, Buro, and Schaeffer (2006) report that with strong evaluations good checker play needs little search depth. Rollouts (Tesauro and Galperin 1997) and Monte Carlo tree search (van Lishout, Chaslot, and Uiterwijk 2007) have also been applied, and backgammon environments are part of general frameworks such as OpenSpiel (Lanctot et al. 2019). The closest work on *variants* concerns the two-player Greek games Portes, Plakoto and Fevga (Papahristou and Refanidis 2011, 2012; Papahristou 2015). Four-player backgammon exists only as informal or commercial designs (Sethares; pat), and the multi-player formats used in practice (chouette, partner games) are team formats rather than free-for-all. Our linear evaluation is closest in spirit to BKG's hand-crafted features.

Multi-player game search. max^n (Luckhardt and Irani 1986) assumes that each player maximises its own payoff, while the paranoid algorithm (Sturtevant and Korf 2000) assumes that all opponents are united against the root player. Best-reply search (Schadd and Winands 2011) lets only the strongest opponent move, and MP-Mix (Zuckerman, Felner, and Kraus 2009) switches strategies depending on the standings. UCT has been analysed for multi-player games (Sturtevant 2008; Kocsis and Szepesvári 2006), and Baier and Kaisers (2020) find that abstracting opponent moves away and focusing on one's own moves can improve multi-player MCTS. These methods are mostly studied on deterministic games; Settlers of Catan (Szita, Chaslot, and Spronck 2010) and Ludo (Alvi and Ahmed 2011; Alhajry, Alvi, and Ahmed 2012) are the nearest stochastic multi-player board games. Ludo's strategies are described with move types such as aggressive, defensive, and fast, which resemble our plan library.

Opponent models inside search. Opponent-model search (M*) (Carmel and Markovitch 1996) replaces the minimax assumption with an explicit model, and probabilistic opponent-model search (PrOM) (Donkers, Uiterwijk, and van den Herik 2001) uses a probability distribution over several models at opponent nodes. Prob-max^n (Sturtevant, Zinkevich, and Bowling 2006) is the closest precedent for our work: it keeps a distribution over candidate opponent models in an n-player game (Spades), updates it from observed play and uses it in max^n search (see also Sturtevant and Bowling 2006). Our work differs in three ways: the game has dice and therefore chance nodes, the observation model is the Boltzmann-rational likelihood from the plan recognition literature, and we measure recognition accuracy separately from playing strength. In Scrabble, Richards and Amir (2007) infer hidden opponent tiles and report a clear gain; we come back to this contrast in the discussion.

Plan and goal recognition. Plan recognition has been studied with plan libraries (Kautz and Allen 1986), Bayesian networks (Charniak and Goldman 1993), and plan grammars (Geib and Goldman 2009); see Sukthankar et al. (2014) and Mirsky, Keren, and Geib (2021) for overviews. Plan recognition as planning (Ramírez and Geffner 2009) and its probabilistic version (Ramírez and Geffner 2010) derive goal likelihoods from a Boltzmann function of cost differences; related cost-based and landmark-based methods exist for navigation and planning domains (Masters and Sardina 2019; Pereira, Oren, and Meneguzzi 2020). Bayesian inverse planning (Baker, Saxe, and Tenenbaum 2009) models people as noisily rational in the same way; the model goes back to Luce's choice axiom (Luce 1959) and appears in games as quantal response (McKelvey and Palfrey 1995). In games, strategy recognition has been applied to discrete build orders in real-time strategy games (Synnaeve and Bessière 2011; Weber and Mateas 2009); our plans are instead playing styles, defined by evaluation weights.

Opponent modelling and type-based reasoning. Surveys of opponent modelling (Albrecht and Stone 2018; Nashed and Zilberstein 2022; van den Herik, Donkers, and Spronck 2005) place our method in the *type-based reasoning* category: the agent keeps a belief over hypothesised types (Albrecht, Crandall, and Ramamoorthy 2016), as in ad-hoc teamwork (Stone et al. 2010) and Bayesian policy reuse (Rosman, Hawasly, and Ramamoorthy 2016). I-POMDPs (Gmytrasiewicz and Doshi 2005) give a principled but intractable formulation of the same idea, and our agent can be seen as a cheap approximation of it. In poker, a Bayesian posterior over opponent strategies combined with a best response (Southey et al. 2005) follows the same two steps as our agent (recognise, then respond); work on poker has also shown the tension between exploiting a model and being robust to modelling errors (Billings et al. 2002; Ganzfried and Sandholm 2011; Johanson, Zinkevich, and Bowling 2008). Learned models (He et al. 2016; Rabinowitz et al. 2018) are an alternative to hand-built libraries. Finally, high-profile multi-player results differ in how much they rely on modelling others: Pluribus reached superhuman six-player poker without opponent modelling (Brown and Sandholm 2019), while in Diplomacy modelling other players is central (Meta Fundamental AI Research Diplomacy Team (FAIR), Bakhtin et al. 2022).

Positioning. We combine these lines of work: plan recognition, with an observation model from the goal recognition literature, is used directly as the opponent model inside a lookahead planner, in a stochastic multi-player game with perfect information. Our evaluation measures recognition accuracy and its effect on decisions and on playing strength separately.



Figure 1: The four-player board. The outer ring shows the 48 points, tinted by quadrant (A: 1–12, B: 13–24, C: 25–36, D: 37–48). Each inner arc is one player's 24-point path from its entry point (dot) to bearing off; the thick end is its home board. A's starting checkers are shown on the ring (2, 5, 3, 5 on A's relative points 1, 12, 17, 19). A's path overlaps D's path (points 1–12) and B's path (13–24) but not C's path.

3 Four-Player Backgammon

Since there is no standard four-player backgammon, we designed a variant with two goals: (i) stay as close as possible to the rules familiar to backgammon players, so that standard intuitions and features carry over, and (ii) make the interaction between players symmetric.

Board and movement. The board is a ring of 48 points divided into four quadrants of 12 points (Figure 1). Players A, B, C and D enter at points 1, 13, 25 and 37 and all move in the same direction, wrapping around from 48 to 1. Each player's path is 24 points long, exactly as in standard backgammon, and its last six points are the player's home board. A path therefore covers the player's own quadrant and the next one. Each player shares 12 points with each of its two neighbours, the one behind it and the one ahead, and never meets the player opposite. Every player is in the same position relative to the others, so the interaction is symmetric.

Setup and turns. Each player starts with the standard 15-checker position on its own path (2 checkers on its 24-point, 5 on the 13-point, 3 on the 8-point and 5 on the 6-point, counted from the end of the path), for a pip count of 167. Players take turns in the order A, B, C, D; a random player starts. Each turn the player rolls two dice, and doubles are played four times.

Standard rules. A point held by two or more checkers of one player is blocked for everyone else. Landing on a single opposing checker (a *blot*) hits it and sends it to its owner's bar. Checkers on the bar must re-enter on the first six points of the owner's path before any other move. Bearing off is allowed only once all of a player's checkers are in its home board, and a higher die may bear off the rearmost checker. A player must use as many dice as possible, and the larger die if only one can be played.

End of game and scoring. Play continues until three players have borne off all their checkers. Places 1st to 4th score 3, 2, 1 and 0 points, so the mean score of a player of average strength is 1.5 and its win rate is 25%. We left out the doubling cube and gammons, both of which would need a separate multi-player design. Games between random players last about 170 turns; games between our informed agents last 104–114 turns.

Variants and testbed. All rule parameters (number of players and points, path length, home size, setup, scoring) can be configured. A *long* preset with 36-point paths makes every pair of players interact, similar to QuadGammon (Sethares), but games take about four times as many turns. The engine is pure Python, separates game, agent and experiment layers, and gives every agent the full observation of each turn (position, dice, legal plays and chosen play), since the game has perfect information. Experiments rotate seats and seed each game and each agent deterministically, so results are identical for any number of worker processes. Like general systems such as Ludii (Piette et al. 2020) and OpenSpiel (Lanctot et al. 2019), the testbed is meant for comparing agents, but it focuses on the multi-player setting that those collections lack for backgammon.

4 Planning with Plan Recognition

Our agent has three components: a feature-based evaluation that also defines a library of plans, a Bayesian recognizer that infers each opponent's plan, and a planner that uses the recognized plans to predict opponent replies.

Position Evaluation and Plan Library

All agents evaluate the position x' reached by a play from the mover's point of view, using a feature vector $\phi(x')$ with nine entries: pip count, checkers borne off, checkers on the bar, made points (two or more checkers), made points in the home board, length of the longest prime, *blot risk* (expected pips lost to hits before the player's next turn, using the classical table of shots per distance and treating the opponents as independent), number of exposed blots, and *hit gain* (pips the play cost opponents by hitting). A plan s is a weight vector w_s, and its value for a play is $V_s = w_s^\top \phi(x')$.

The plan library contains five styles (Table 1). balanced is a hand-tuned general-purpose evaluation; race ignores structure and hits and only advances; aggressive values hitting highly and accepts more risk; blocker values made points and primes; safe strongly avoids exposed blots. A sixth hypothesis, random, chooses uniformly among legal plays. The greedy *heuristic* agent plays the best play under balanced and is our main baseline. Heuristic agents with the other styles act as opponents whose plan is known.

plan	pips	off	bar	made	home	prime	risk	blots	hit
balanced	-1	.5	-4	1	1.5	1	-1	$-.3$	.35
race	-1	1	-4	.2	0	0	$-.2$	0	.05
aggressive	-1	.5	-2	.3	.5	.3	$-.2$	0	1.5
blocker	-1	.5	-4	2.5	3	3	$-.6$	$-.2$	.2
safe	-1	.5	-4	1	.5	.3	-3	-1.5	.2

Table 1: The plan library: weight vectors over the nine evaluation features. random (uniform play) is a sixth hypothesis.

Plan Recognition

The agent assumes that each opponent q follows one fixed, hidden plan $s \in \mathcal{S} = \{\text{balanced}, \text{race}, \text{aggressive}, \text{blocker}, \text{safe}, \text{random}\}$ for the whole game. Following Bayesian inverse planning (Baker, Saxe, and Tenenbaum 2009) and probabilistic plan recognition as planning (Ramírez and Geffner 2010), an opponent with plan s is modelled as Boltzmann-rational. When q faces position x with dice d and legal plays $\Pi(x, d)$, $|\Pi| = n \geq 2$, the value of each play is first standardised over that turn's legal plays:

$$z_s(\pi) = \frac{V_s(\pi) - \max_{\pi'} V_s(\pi')}{\max(\sigma_s, 1)}, \tag{1}$$

where σ_s is the standard deviation of V_s over $\Pi(x, d)$. Plans differ greatly in the scale of their values, and the standardisation lets one inverse temperature β work for all of them; the floor of 1 avoids inflating differences between almost equivalent plays. The choice model is

$$P(\pi \mid s) = \frac{\exp(\beta\, z_s(\pi))}{\sum_{\pi' \in \Pi} \exp(\beta\, z_s(\pi'))}, \qquad P(\pi \mid \text{random}) = \frac{1}{n}. \tag{2}$$

To stay robust to opponents outside the library, we mix in a lapse rate ε: $P_\varepsilon(\pi \mid s) = \varepsilon/n + (1 - \varepsilon)P(\pi \mid s)$. Starting from a uniform prior $b_q^0(s) = 1/6$, every observed decision of q with play π_t updates the posterior by Bayes' rule,

$$b_q^t(s) \propto b_q^{t-1}(s)\, P_\varepsilon(\pi_t \mid s), \tag{3}$$

which we compute in log space. Turns with a single legal play carry no information and are skipped. We use $\beta = 10$ and $\varepsilon = 0.05$. A large β makes the likelihood sharp, so that a few characteristic decisions (an aggressive player hitting when it need not, a safe player refusing to leave a blot) quickly separate the plans. The lapse rate keeps one surprising move from eliminating the true plan. The library-plus-lapse construction is the standard type-based approach, and its behaviour when the true type is not in the library is analysed by Albrecht, Crandall, and Ramamoorthy (2016). Recognition costs one feature evaluation per legal play and plan for each observed turn.

Planning with Predicted Opponents

The planner (Algorithm 1) refines the greedy balanced evaluation by looking one opponent turn ahead, but only where an opponent can affect it directly, by hitting one of its blots. It first ranks all legal plays by V_{balanced} and keeps the top $k = 5$ candidates. For a candidate leading to x', let $Q(x')$ be the opponents that have a direct or indirect shot at one of the agent's blots in x'; in our variant these can only be the two neighbours. If $Q(x') = \emptyset$, the candidate keeps its static value. Otherwise, the generic blot-risk term is computed only for the other opponents, and for each $q \in Q(x')$ the planner adds the expected change in value from q's next turn:

$$\Delta_q(x') = \sum_r P(r) \sum_s \hat{b}_q(s) \sum_{\rho \in H_q(x', r)} P(\rho \mid s) \left[V^0(x' \oplus \rho) - V^0(x')\right]. \tag{4}$$

Here r ranges over the 21 distinct rolls with probability $P(r)$, $H_q(x', r)$ are q's legal replies that hit one of our checkers, $P(\rho \mid s)$ is the Boltzmann policy of plan s evaluated from q's point of view, V^0 is the balanced value without any blot-risk term, and $\hat{b}_q$ is q's posterior, truncated to plans with mass at least 0.02 and renormalised. Rolls that cannot reach any blot are skipped. The candidate's value is $U(x') = w_{\text{balanced}}^\top \phi_{-Q}(x') + \sum_{q \in Q(x')} \Delta_q(x')$, and the agent plays the candidate with the highest U.

This is a selective expectimax of depth two (our play, then a chance node and an opponent decision) in which the opponent node is a posterior-weighted mixture of plan policies rather than a max or min node, similar to PrOM search (Donkers, Uiterwijk, and van den Herik 2001) and Prob-max^n (Sturtevant, Zinkevich, and Bowling 2006). Each threatening opponent is treated independently. The approach contains two approximations: replies that do not hit count as neutral, and the expansion stops after one opponent turn.

Algorithm 1 Planner decision with plan recognition

Require: position x, dice d, legal plays Π, posteriors b_q
$C \leftarrow$ top-k plays of Π by V_{balanced}
for each $\pi \in C$ with successor x' **do**
  $Q \leftarrow$ opponents with a shot at a blot in x'
  $U(\pi) \leftarrow w_{\text{balanced}}^\top \phi_{-Q}(x')$ {risk from others only}
  for each $q \in Q$ and roll r that reaches a blot **do**
    $H \leftarrow$ replies of q to r that hit us
    for each plan s with $\hat{b}_q(s) \geq 0.02$ **do**
      $U(\pi)$ += $P(r)\, \hat{b}_q(s) \sum_{\rho \in H} P(\rho \mid s)\, [V^0(x' \oplus \rho) - V^0(x')]$
    end for
  end for
end for
return $\arg\max_{\pi \in C} U(\pi)$

Cost. The planner takes 24–28 ms per decision compared with 0.24–0.28 ms for the greedy heuristic, about a hundred-fold increase, which is still fast enough to run thousands of games. Recognition makes the planner *cheaper*: once the posterior is concentrated, few plans pass the 0.02 threshold, and only their policies have to be evaluated in Eq. (4).

Ablation and Oracle

To separate the contribution of recognition from that of lookahead, the *ablation* (`recognize=False`) is the same

planner whose opponent model stays at the uniform prior over all six plans: it still predicts replies, but with a fixed mixture of all plans. For the analysis added for this paper, the *oracle* planner is the same planner whose model of each opponent is that opponent's true plan, with probability 1. The ablation gives a lower bound and the oracle an upper bound on what plan recognition can contribute within this planner.

5 Experimental Setup

Four agents take part: the planner, the ablation, the greedy heuristic, and uniformly random play. All results come from the evaluation script of our testbed (master seed 2026), which runs five experiments in about 58 CPU-minutes (145 s on 24 cores). The decision-level analysis and the oracle come from a supplementary script that replays the same games with the same seeds; it reproduces every score reported by the main script exactly.

Protocol. The line-up is rotated by one seat every game to cancel seat effects. In experiments where one agent plays against several copies of another, every seat counts as an appearance, so the three heuristic copies in Experiment 2 together have 2,400 appearances. Wherever two agents are compared in separate runs (planner and ablation against heuristics, or each agent against the mixed opponents), the runs use the same game and agent seeds, so the comparisons are paired: the dice are the same, and the games differ only where the agents' decisions differ. Planner agents use no randomness.

Metrics and statistics. We report the win rate (share of 1st places) with 95% Wilson intervals (Wilson 1927) and the mean placement score with 95% normal intervals. Parity is 25% and 1.5. For comparisons between agents we report paired per-game score differences with normal 95% intervals and two-sided p-values.

Experiments. (1) Each informed agent against three random agents (400 games each). (2) Planner, and separately the ablation, against three heuristics (800 games each). (3) A round robin of planner, ablation, heuristic and random at one table, in all six cyclic seatings (96 games each, 576 games). Every agent therefore has every other agent as left and right neighbour equally often. (4) Planner, ablation, heuristic and oracle, each against race, aggressive and random opponents in all six seatings (384 games each). (5) Recognition accuracy: the planner observes three opponents whose plans are drawn uniformly from the six plans (240 games, 720 identifications).

6 Results

Table 2 and Figure 2 summarise playing strength, Table 3 gives the paired comparisons, and Table 4 reports what the planner's components change at the level of individual decisions.

(1) Sanity check against random play. All informed agents win more than 90% of their games against three random players and almost never finish worse than second. The heuristic finished third 3 times, the planners never. Against random play the planners hit more often (8.1 vs. 7.1 hits per game) without being hit more. The differences between the informed agents are too small to rank them here, which confirms that random opponents are not a useful benchmark for this question.

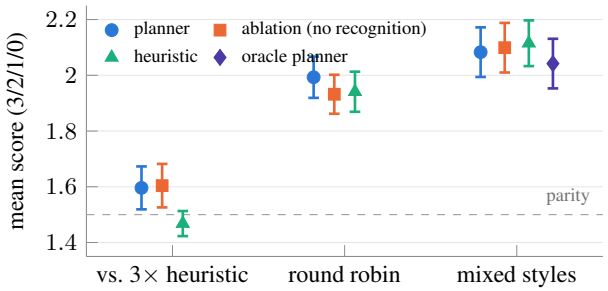


Figure 2: Mean placement score with 95% intervals in the three experiments against informed opponents. The planner and the ablation clearly beat the heuristic opponents in (2). In (3) and (4) all intervals overlap.

(2) Lookahead beats the greedy evaluation. Against three copies of the heuristic it is built on, the planner wins 28.9% of games, and the Wilson interval [25.8, 32.1] lies above parity. Its mean score of 1.60 is higher than the heuristics' 1.47, and the paired per-game margin over the mean of the three heuristics at the same table is $+0.128 \pm 0.103$ ($p = 0.015$). The only difference between the two agents is the lookahead over opponent replies, which changes 10.2% of the planner's decisions (Table 4). The ablation scores the same: it wins exactly the same number of games (231), and the paired score difference is -0.008 ± 0.053 ($p = 0.78$). This result was expected: all three opponents play balanced, the style the uniform prior already contains, and the recognizer's posterior mostly converges to balanced or to the similar blocker and safe plans.

(3) Round robin. With all four agents at one table, the planner has the highest win rate (36.5% against 29.7% for the ablation and 33.5% for the heuristic) and the highest mean score (1.99 against 1.93 and 1.94). The paired score difference between planner and ablation, $+0.061 \pm 0.123$, is not significant ($p = 0.33$). The paired win-rate difference, $+6.8$ percentage points, reaches $p = 0.045$ without correction, but it is one of eight recognition-related comparisons in Table 3 and does not survive a Bonferroni correction (threshold $0.05/8 \approx 0.006$). The planner's higher win rate is also offset by fewer second places (176 vs. 215), so its placements are shifted rather than improved overall. We therefore report this as a suggestive trend rather than a finding.

(4) Mixed opponent styles. Against a race, an aggressive and a random opponent, the setting where knowing each opponent's plan *should* matter most, the three agents are statistically indistinguishable (planner 2.08, ablation 2.10, heuristic 2.11; all paired $p > 0.5$). The opponent model does change behaviour: the planner is hit 13% less often than the ablation (1.88 vs. 2.16 times per game) and 16% less often than the heuristic, because it correctly expects the

experiment	agent	n	win rate [95% CI]	mean score	hits made	hits taken	ms/decision
(1) vs. 3× random	planner	400	92.0% [88.9, 94.3]	$2.92 \pm .03$	8.06	1.37	27.7
	ablation	400	93.5% [90.6, 95.5]	$2.94 \pm .02$	8.09	1.45	58.6
	heuristic	400	90.5% [87.2, 93.0]	$2.90 \pm .03$	7.15	1.37	0.28
(2) vs. 3× heuristic	planner	800	28.9% [25.8, 32.1]	$1.60 \pm .08$	1.42	1.25	24.3
	ablation	800	28.9% [25.8, 32.1]	$1.60 \pm .08$	1.46	1.28	25.3
	(heuristic opponents)	2400	23.7% [22.0, 25.5]	$1.47 \pm .04$	1.32	1.38	0.24
(3) round robin	planner	576	**36.5%** [32.6, 40.5]	$\mathbf{1.99} \pm .07$	3.77	1.44	26.1
	ablation	576	29.7% [26.1, 33.5]	$1.93 \pm .07$	3.83	1.49	33.5
	heuristic	576	33.5% [29.8, 37.5]	$1.94 \pm .07$	3.18	1.51	0.26
	random	576	0.3% [0.1, 1.3]	$0.13 \pm .04$	1.31	7.63	0.003
(4) vs. race, aggr., random	planner	384	41.1% [36.3, 46.1]	$2.08 \pm .09$	4.05	**1.88**	27.0
	ablation	384	41.9% [37.1, 46.9]	$2.10 \pm .09$	4.35	2.16	36.0
	heuristic	384	38.5% [33.8, 43.5]	$2.11 \pm .08$	3.84	2.24	0.24
	oracle planner	384	37.8% [33.0, 42.7]	$2.04 \pm .09$	–	–	–

Table 2: Playing strength. Parity is a 25% win rate and a mean score of 1.5. Scores are means with 95% interval half-widths; hits are per game. In each experiment the agents are compared on the same seeds (paired). The oracle row comes from the supplementary replay, which did not log per-game hit and timing statistics.

comparison (per-game, paired)	diff.	95% CI	p
(2) planner − ablation, score	−0.008	±0.053	0.78
(2) planner − mean heuristic, score	+0.128	±0.103	0.015
(3) planner − ablation, score	+0.061	±0.123	0.33
(3) planner − ablation, win	+0.068	±0.066	0.045
(3) planner − heuristic, score	+0.052	±0.125	0.41
(4) planner − ablation, score	−0.016	±0.074	0.68
(4) planner − heuristic, score	−0.031	±0.095	0.52
(4) oracle − planner, score	−0.042	±0.062	0.19
(4) oracle − ablation, score	−0.057	±0.085	0.18

Table 3: Paired comparisons. In (3) the agents play in the same games; in (2) and (4) they play on identical seeds. Only the lookahead margin in (2) remains significant after correcting for the eight recognition-related tests.

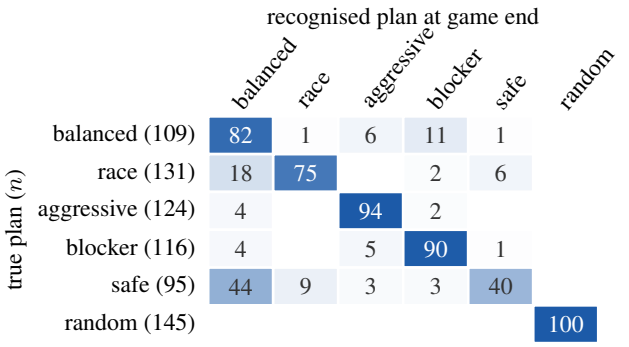


Figure 3: Plan recognition confusion matrix (percent of each true plan; n opponents per row; 720 in total). The main error is safe recognised as balanced.

aggressive neighbour to hit and the race neighbour not to. This safety costs some tempo and hitting chances of its own (4.05 vs. 4.35 hits made), and the score is unchanged. Most strikingly, the *oracle* planner, which knows every opponent's plan from the first move, scores 2.04, no better than the planner with inferred plans (-0.042 ± 0.062, $p = 0.19$). Better recognition cannot help this planner.

(5) Plan recognition accuracy. Against opponents with known plans, the most likely plan at the end of the game is correct for 81.9% of the 720 opponents, compared with 16.7% chance (Figure 3). Recall is highest for random (100%), aggressive (94%) and blocker (90%), which show characteristic decisions, followed by balanced (82%) and race (75%). safe is the hard case at 40%: it is mistaken for balanced in 42 of 95 games, because the two plans share most weights and differ mainly in how much they avoid blots, which only matters in positions where both choices are available. Recognition is fast (Figure 4): 35% after one observed decision, 50% after three, 69% after ten and 84% after twenty, with 22 observed decisions per opponent and game on average.

7 Why Accurate Recognition Does Not Help

Recognition is accurate and fast, and its predictions change the planner's behaviour in measurable ways (fewer hits taken), yet scores do not improve, even with a perfect model. The decision logs in Table 4 and the experiments above point to four reasons.

The model is used often but rarely decides. The opponent model enters at least one candidate's value in 75–81% of decisions, but it changes the chosen play in only 2.8–5.6% of all decisions, and in fewer than 7% even of the decisions where it is used. Lookahead itself changes two to three times as many decisions (10–14%). With about 21 decisions per game, recognition changes one or two moves per game, and those moves are mostly close calls between candidates of similar value. A difference of 0.06 points in mean score would need several thousand games per condition to detect,

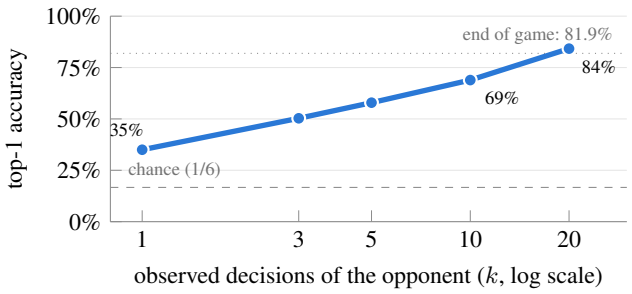


Figure 4: Recognition accuracy after the first k observed decisions of an opponent ($n = 720$ for $k \leq 10$, $n = 505$ for $k = 20$; few opponents make 40 decisions).

	(2) vs. heur.	(3) r. robin	(4) mixed
planner decisions	16,144	11,923	8,019
model consulted	74.8%	79.8%	81.2%
lookahead ≠ greedy	10.2%	13.3%	13.5%
recognition ≠ ablation	**2.8%**	**4.9%**	**5.6%**
of decisions where consulted	3.7%	6.1%	6.9%

Table 4: Decision-level analysis. At every planner decision we also compute the greedy balanced choice and the choice the ablation would make in the same position. "Model consulted": at least one candidate leaves a blot exposed to an opponent, so that Eq. (4) is evaluated.

and our experiments are designed for effects about twice that size.

The model only enters through hit risk. The planner uses the posterior for one purpose: estimating how likely and how costly it is to be hit in the next turn. Everything else about an opponent's plan is ignored: whether a race opponent is about to leave the shared quadrant, whether a blocker opponent is building a prime in front of our back checkers, or which neighbour is the leader to attack. With the narrow channel the planner uses, recognizing a plan can at best shift risk estimates. In Experiment 4 it does exactly that (13% fewer hits taken), and the safety gained there is offset by lost tempo.

Confusable plans make the same predictions. Recognition errors are concentrated where they do not matter. safe and balanced differ mostly in blot avoidance, and in the hitting decisions that Eq. (4) cares about they make nearly the same choices. Any opponent will usually hit a blot if it can, so the predicted hit probabilities of most plans are similar. This is the mirror image of the robustness argument of Johanson, Zinkevich, and Bowling (2008): a model that is wrong in ways that do not change the best response costs little, and a correct model gains little for the same reason. The oracle result confirms that the remaining errors cost nothing.

Shallow lookahead limits exploitation. The planner looks one opponent turn ahead and treats non-hitting replies as neutral. Consistent with Hauk, Buro, and Schaeffer (2006), a reasonable evaluation already captures most of what shallow search can add in backgammon. Exploiting a recognised plan strategically, for example by racing against a race player or staying back against a blocker, takes plans that span many turns, and a one-turn horizon cannot represent them. Baier and Kaisers (2020) reach a similar conclusion for multi-player MCTS: improving one's own move selection pays off more than modelling opponents. Pluribus (Brown and Sandholm 2019) is a prominent case of strong multi-player play without any opponent modelling. The contrast with Scrabble (Richards and Amir 2007) is informative: there, inference recovers *hidden* information (the opponent's tiles) that the player otherwise lacks, while backgammon is fully observable, and the opponent's plan is only useful to the extent that it changes the opponent's predicted moves.

Neighbour structure. In our variant only the two neighbours can ever hit a player, and the opposite player is irrelevant to the hit-risk term. Half of the recognised models are therefore never used for prediction at a given moment, and the random opponent, which is recognised perfectly, hardly threatens anyone. The *long* variant, in which all players interact, would give the model more to do.

Cost and benefit. The planner is about a hundred times slower than the greedy heuristic and is clearly stronger against it, so lookahead pays for its cost. Recognition is cheap and, through posterior truncation, even reduces planning time (26 vs. 34 ms in the round robin, 28 vs. 59 ms against random players). It does no harm, but in the current planner it does not improve play.

8 Limitations

Our plans are hand-crafted evaluation styles and the opponents in the recognition experiment play these styles exactly, which favours recognition; real opponents would follow plans outside the library, change plans during a game, or adapt to the agent. All agents share one feature set, so the planner's search and the opponents' styles are aligned in a way that learned or human opponents would not be. Our sample sizes detect differences of about 0.1 in mean score; smaller effects of recognition, such as the trend in the round robin, cannot be ruled out. Finally, we evaluated only the default 24-point variant with four players.

9 Conclusion and Future Work

We introduced a free-for-all four-player backgammon variant with an open-source testbed, and an agent that uses Bayesian plan recognition as the opponent model inside a selective expectimax planner. Lookahead clearly improves on the greedy evaluation it is built on. Plan recognition identifies opponents' plans reliably (82% top-1 accuracy, 69% within ten decisions), but it changes few decisions and does not improve play significantly, and neither does an oracle with perfect knowledge of the plans. This is a useful lesson for opponent modelling in multi-player games: *accurate models only help if the planner can act on them*, and recognition accuracy alone does not show whether an opponent

model is useful.

Our results point to several directions for future work. First, the opponent model should enter more of the evaluation, for example by predicting opponents' blocking and racing behaviour and not only their hits, and the search should go deeper, through 2-ply expectimax or Monte Carlo rollouts (Tesauro and Galperin 1997) that use the recognised policies instead of a fixed base policy. Second, the recognised plans can be used strategically, e.g., to choose which neighbour to attack or when to race, in the spirit of MP-Mix (Zuckerman, Felner, and Kraus 2009). Third, the plan library can be larger or learned (He et al. 2016; Rabinowitz et al. 2018), plans can change during a game, and recognition can be made open-world for opponents outside the library (Albrecht, Crandall, and Ramamoorthy 2016). Fourth, a learned evaluation trained by self-play (Tesauro 1995) would give a stronger base agent. Finally, the *long* variant, other numbers of players, the doubling cube, and coalition and kingmaking effects (Meta Fundamental AI Research Diplomacy Team (FAIR), Bakhtin et al. 2022), together with games against human players and larger samples, would show whether the small effects seen in the round robin are real.

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